Hierarchical Input-Process-Output Model

This section presents the Hierarchy plus Input-Process-Output (HIPO) model of theWebXR-Enabled Digital Heritage Platform for the NHCP Museo ng Libingan sa Ilalim ng Lupa ng Nagcarlan. The model organizes the system from its principal function into progressively detailed visitor, account, participation, and administration modules.

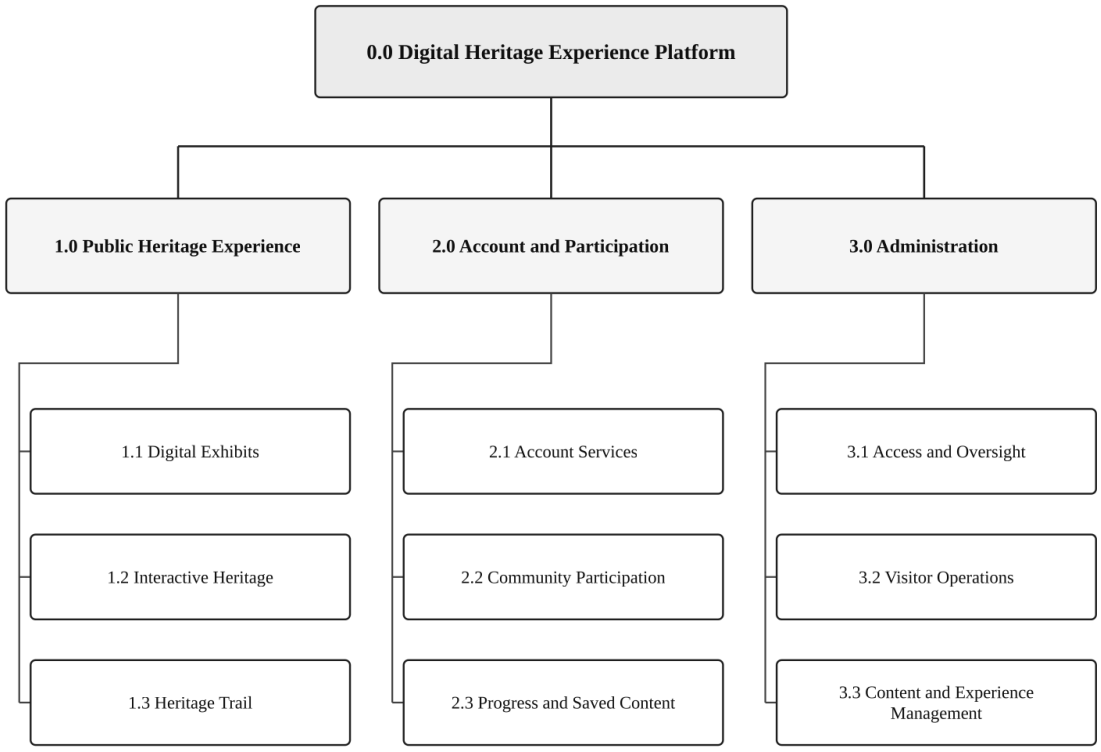


Figure A56. System-Level HIPO Visual Table of Contents.

Figure A56 shows the top-down functional decomposition of the platform into public heritage experiences, account and participation services, and administration functions.

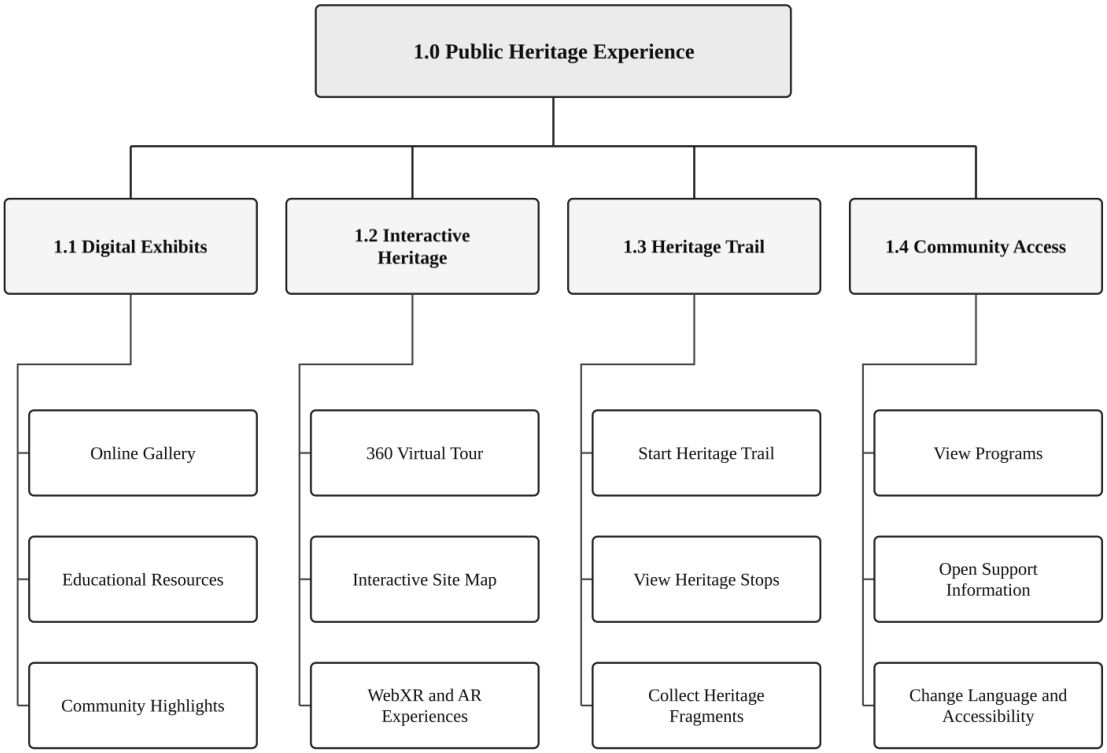


Figure A57. Public Visitor HIPO Hierarchy.

Figure A57 presents the public-facing heritage functions available for viewing collections,opening immersive experiences, following the heritage trail, and accessing visitor information.

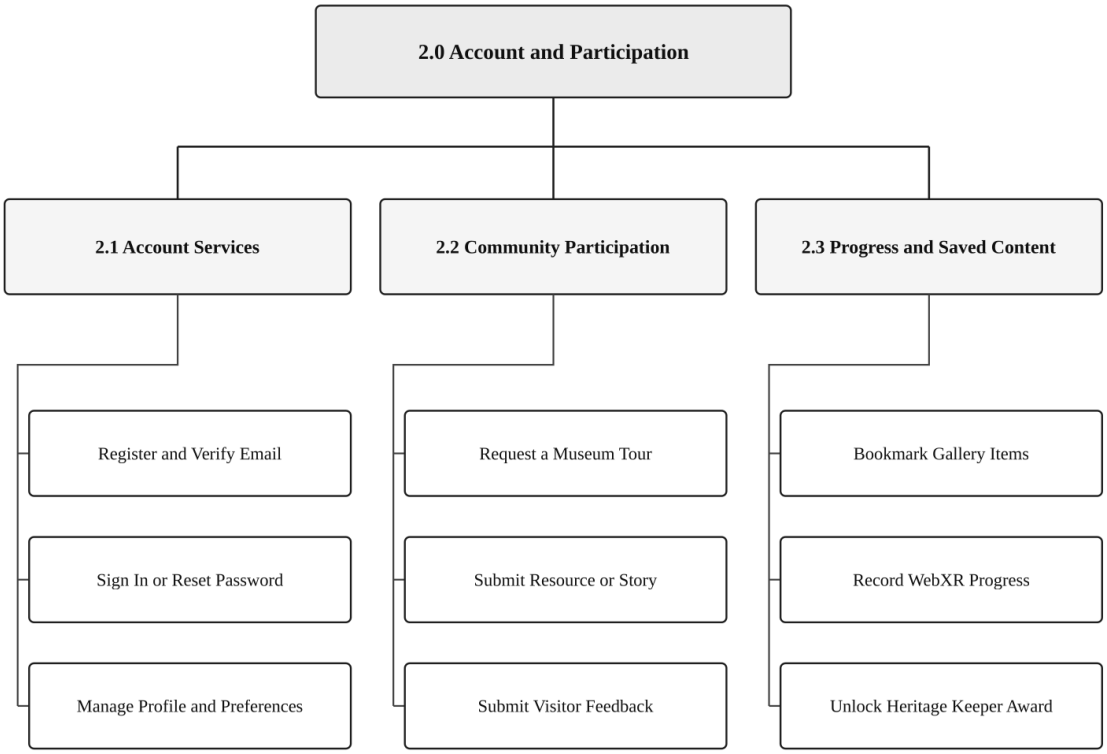


Figure A58. Public Account and Engagement HIPO Hierarchy.

Figure A58 shows the account, community participation, bookmarking, WebXR progress, andHeritage Keeper functions available to public users.

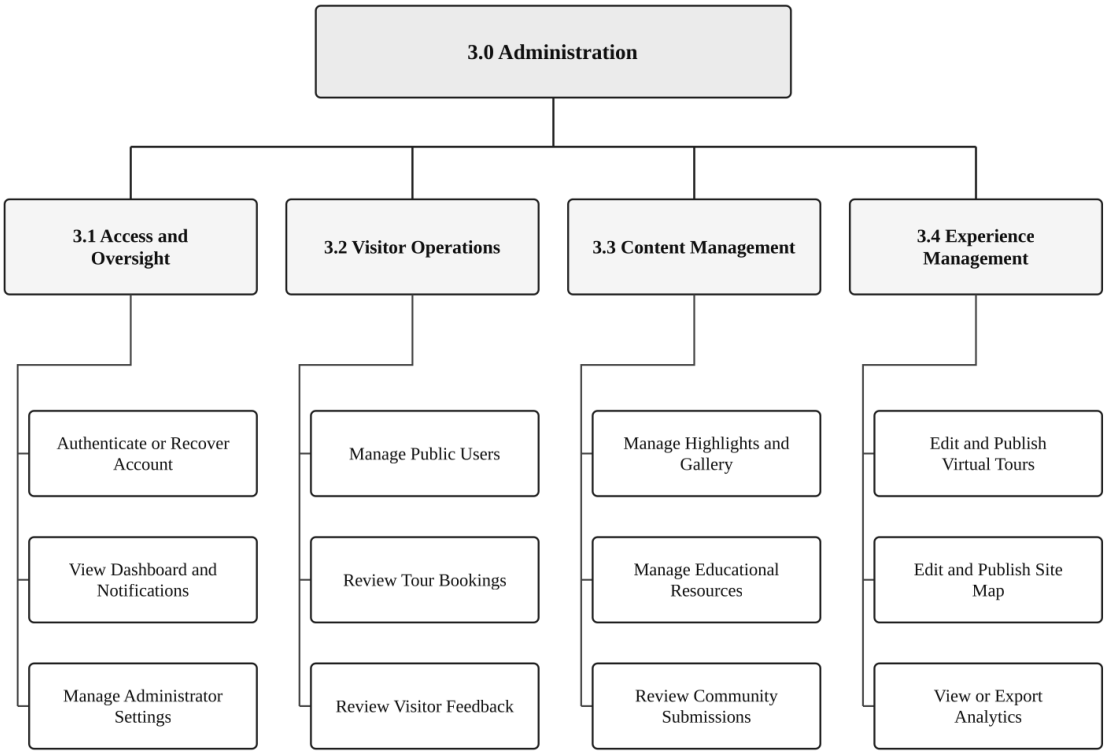


Figure A59. Administrator HIPO Hierarchy.

Figure A59 shows the administrator portal hierarchy for secure access, visitor operations,heritage content management, immersive-experience editing, and analytics reporting.

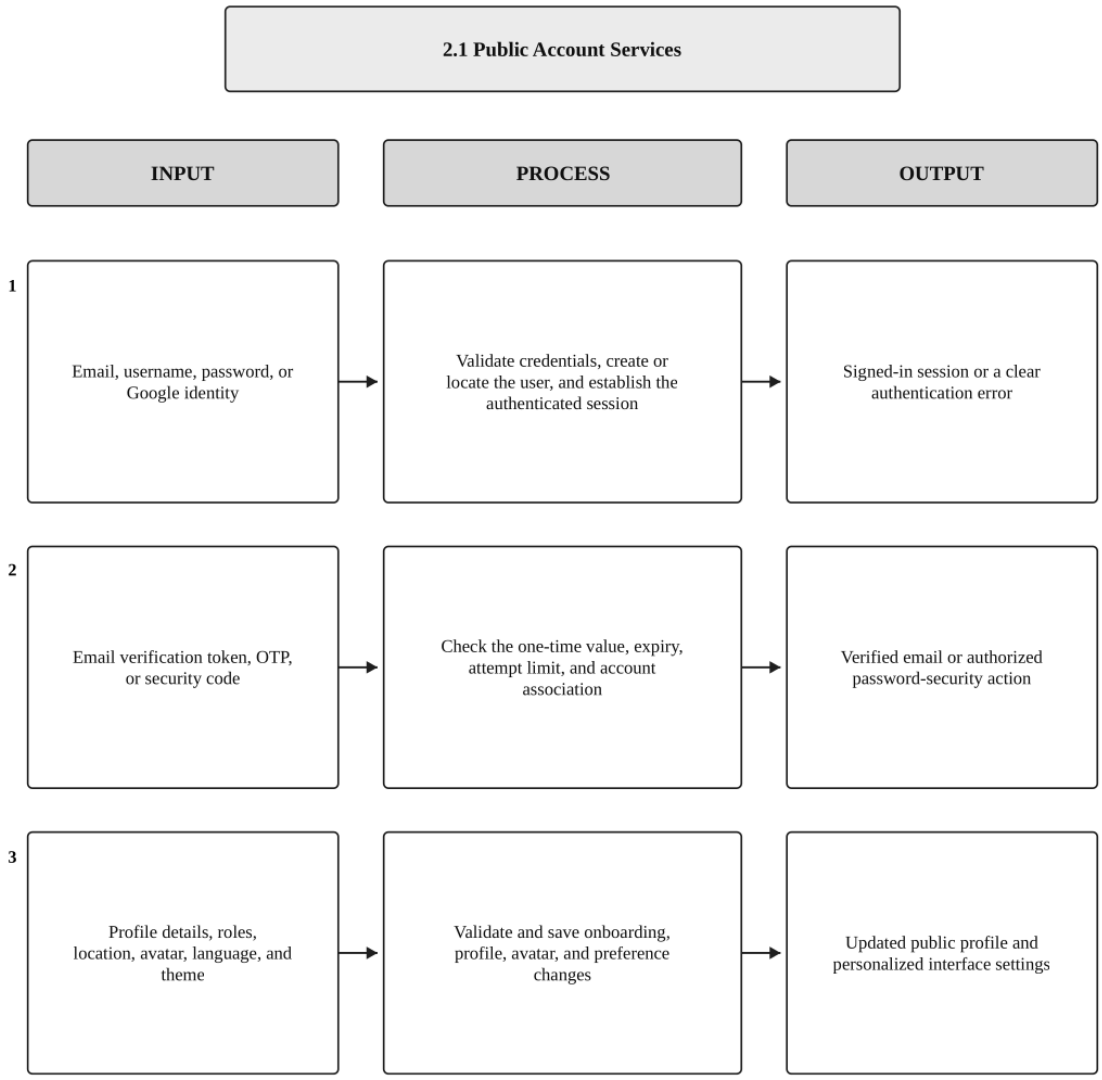


Figure A60. Public Account Services IPO Overview.

Figure A60 identifies the inputs, processing, and outputs used to register, authenticate,verify, secure, and personalize a public account.

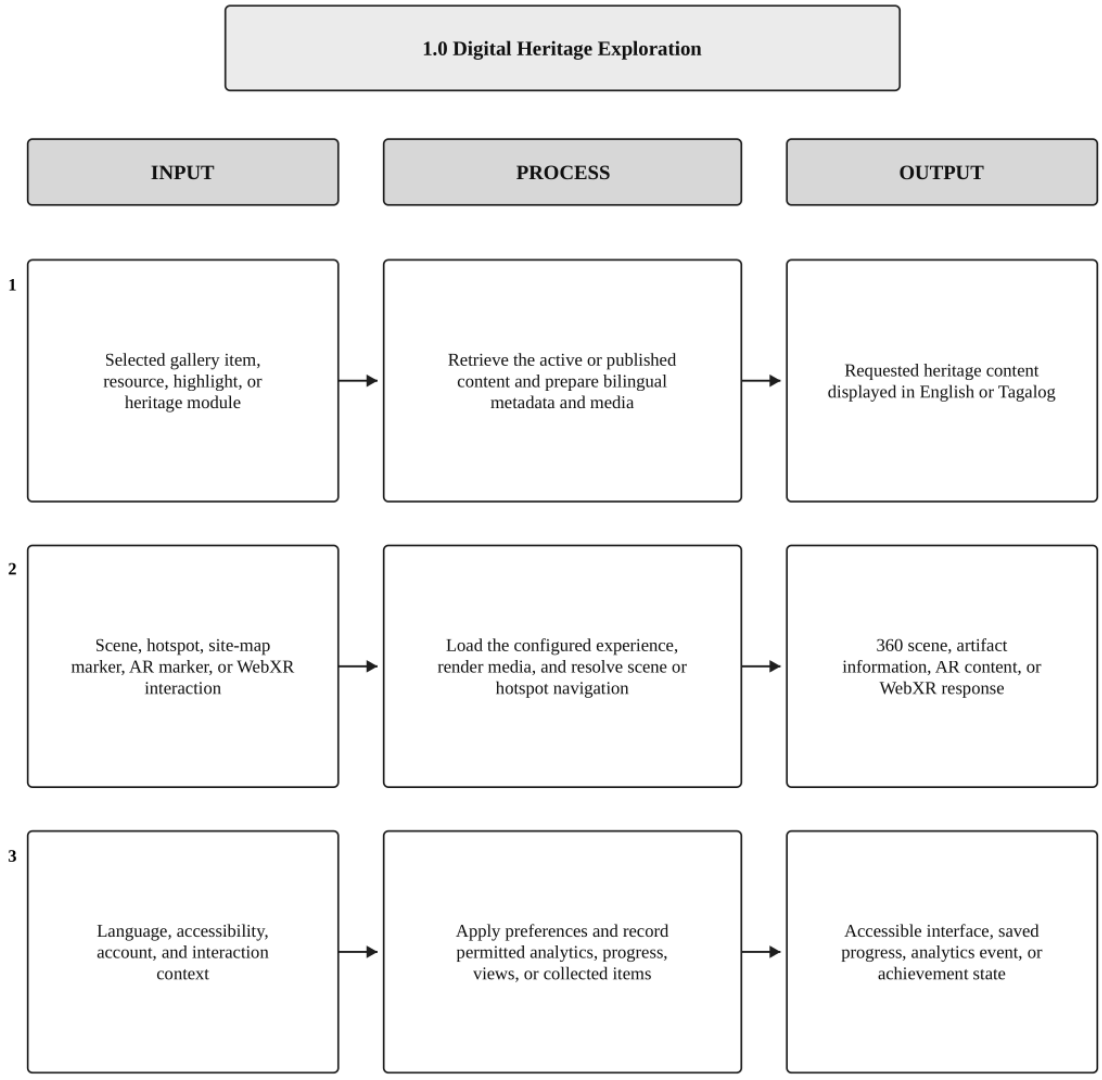


Figure A61. Digital Heritage Exploration IPO Overview.

Figure A61 shows how visitor selections and immersive interactions are processed into bilingual exhibits, 360-degree navigation, augmented-reality content, WebXR responses, and saved progress.

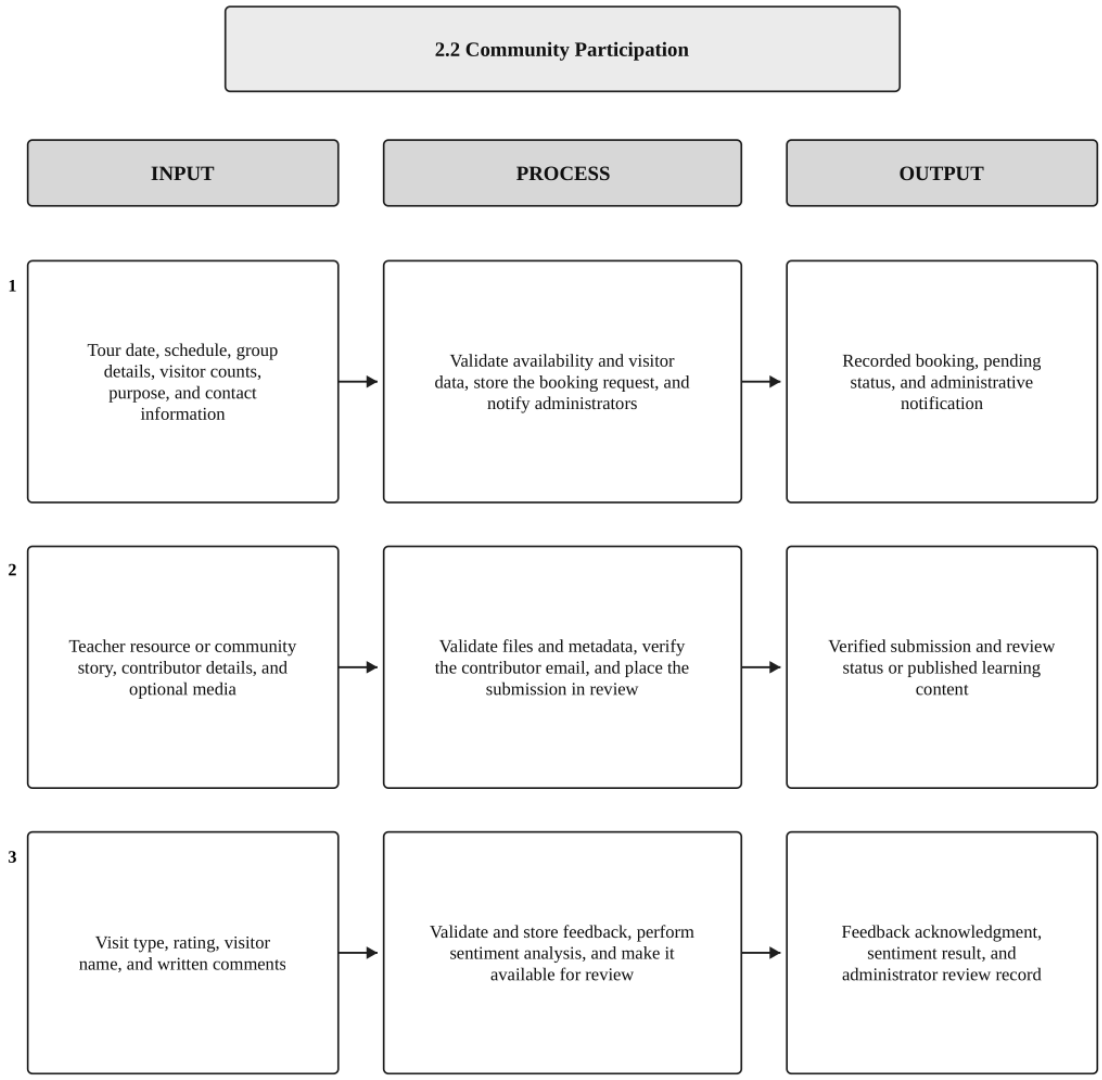


Figure A62. Community Participation IPO Overview.

Figure A62 presents the input, processing, and output relationships for tour requests, community-contributed materials, and visitor feedback.

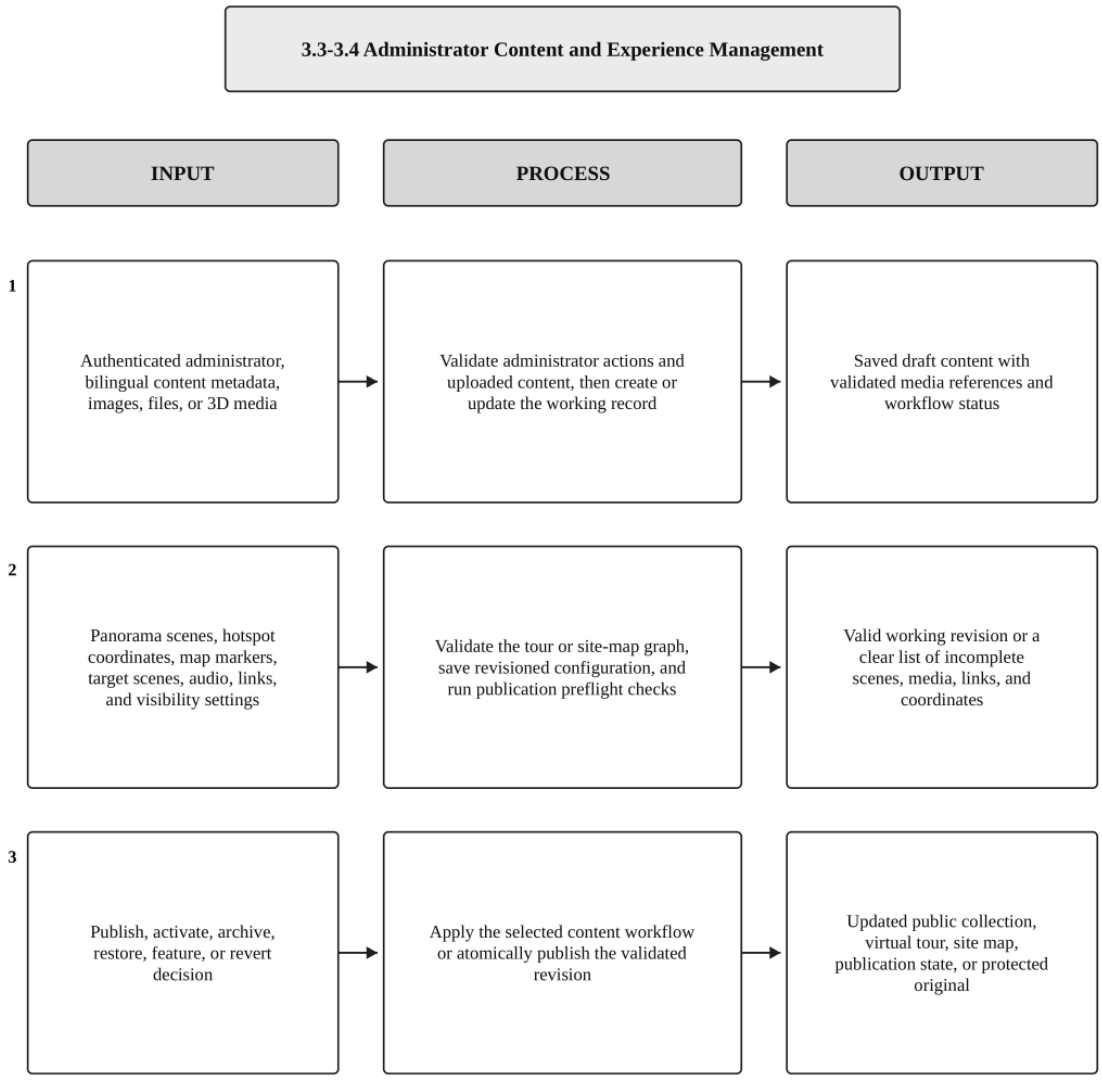


Figure A63. Administrator Content and Experience Management IPO Overview.

Figure A63 shows how administrators transform bilingual metadata, multimedia assets, scenes,hotspots, and publication decisions into validated public content and revisioned immersive experiences.

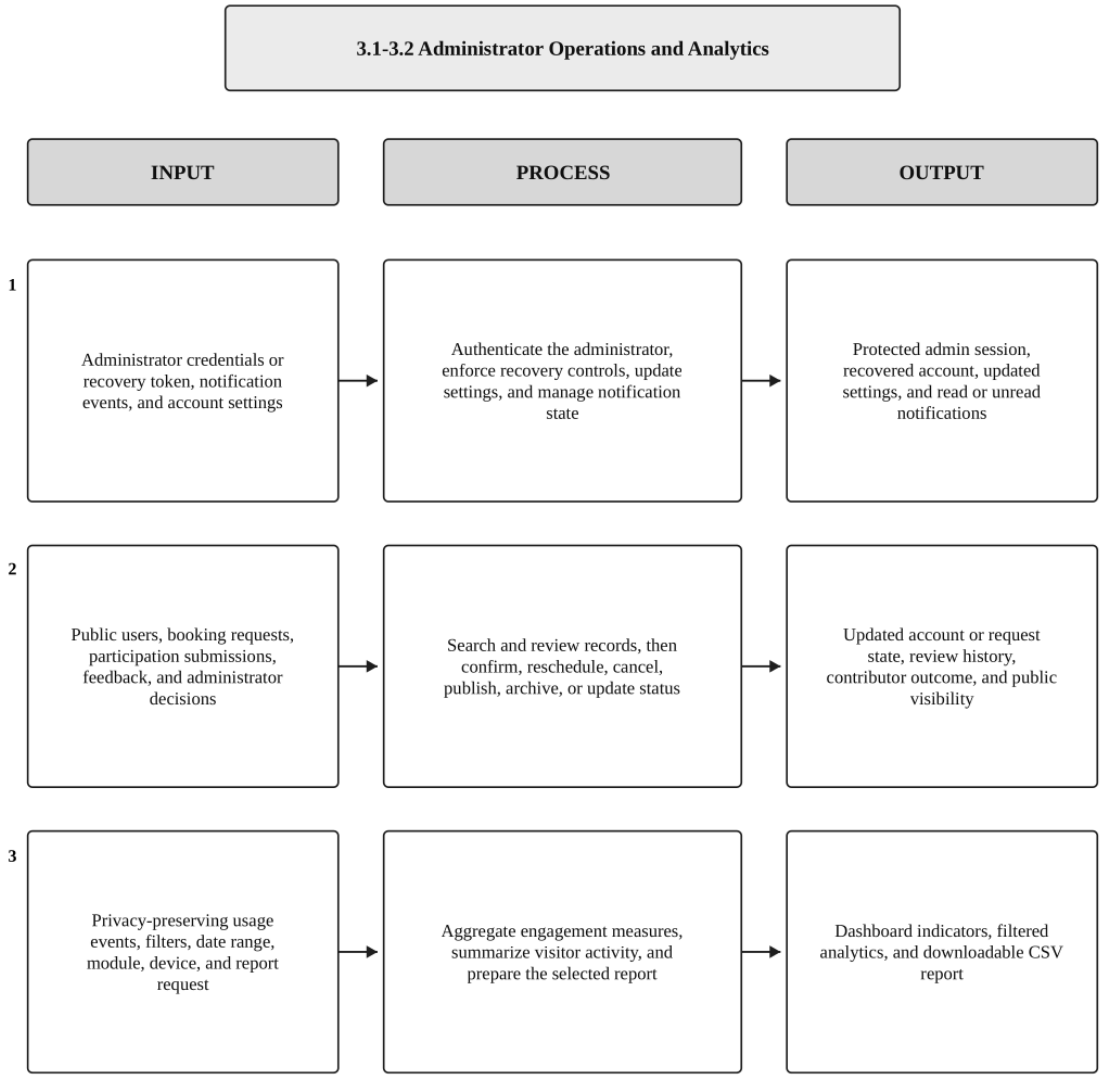


Figure A64. Administrator Operations and Analytics IPO Overview.

Figure A64 presents the inputs, processing, and outputs for administrator access, operational review queues, notifications, booking decisions, and analytics reporting.